

Highlights

Motion-Conditioned Topological Artificial Intelligence for Fragmentation-Risk Forecasting in Unmanned Aerial Vehicle Networks

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- Fragmentation transitions are scored with one-to-one event matching.
- Motion, topology, and graph sources are isolated in a full factorial.
- Twenty seeds and six horizons test the primary forecasting claim.
- Measured three-UAV UWB topology provides direct multi-UAV transfer evidence.
- Packet, latency, and physical relay constraints bound operational claims.

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Abstract

Rapid motion and correlated radio variation can partition a flying ad hoc network before a reactive connectivity monitor leaves enough time for intervention. This paper presents Kinetic-TopoGuard, a short-horizon predictor of the future number of connected components and of connected-to-fragmented transition risk. The method combines the current graph state, velocity-projected link margins, zero-dimensional persistence images, graph summaries, residual regression, and validation-selected risk thresholds. Evaluation separates sample-level fragmentation classification from one-to-one fragmentation-event precision, recall, and F_1 . A run-grouped 20-seed simulation is complemented by a five-seed neural baseline extension, a six-horizon sweep, a full 2^3 feature-source factorial, a load-swept SimPy packet experiment, and complete host-loop latency measurement. Three real-data layers test different parts of the evidence chain: public field trajectories, measured UAV-to-UAV 60 GHz links with an explicitly transported sensitivity analysis, and MILUV motion-capture plus inter-robot UWB first-path-power topology. AERPAAW cellular traces increase time-held-out LTE link-state F_1 from 0.828/0.679 to 0.961/0.980, while the temporally robust throughput tests are negative and are reported as such. The 60 GHz link model increases held-out link-viability F_1 from 0.731 to 0.923. At the 0.6 s horizon, Kinetic-TopoGuard does not improve count MAE over persistence (0.133 versus 0.132), but it yields event $F_1 = 0.483$ with 3.53 false events/minute, compared with 0.414 and 13.24 for shallow ML. MILUV transfer exposes substantial domain shift: Kinetic-TopoGuard event F_1 is only 0.08 at the primary threshold. The study supports reproducible short-horizon warning research, but it does not claim a live digital twin, measured outdoor FANET IP packet delivery, or field deployment.

Keywords: Flying ad hoc networks, Persistent homology, Fragmentation forecasting, Unmanned aerial vehicles, Event detection, Engineering decision support

1. Introduction

Flying ad hoc networks (FANETs) support cooperative sensing, mapping, surveillance, and emergency response without requiring fixed communication infrastructure. Their topology changes more rapidly than that of terrestrial mobile networks because aircraft move in three dimensions, execute mission turns, and can separate at tens of metres per second [1, 2]. A controller that learns about a partition only after links disappear is therefore reactive. A useful predictor must estimate whether the currently connected swarm will fragment at a future control horizon, issue a warning before the transition, and expose enough structure for an engineer to assess false-alert cost.

The zeroth Betti number, β_0 , is the number of connected components. It gives a direct network interpretation: $\beta_0 = 1$ denotes a connected graph and $\beta_0 > 1$ denotes fragmentation. Exact graph traversal or disjoint-set algorithms compute β_0 for an observed graph [3, 4], but they do not forecast its future value. Persistent homology complements a single-radius graph by recording how components merge as the distance threshold grows. A persistence image turns those merge scales into a stable fixed-dimensional feature [5]. In a mobile network, however, topology alone is insufficient: velocity determines whether a marginal link is closing or opening. This motivates a motion-conditioned topological forecaster.

Graph neural networks (GNNs) have been applied to routing, clustering, and coverage [6, 7, 8], while topology-aware networks combine message passing with persistent-homology descriptors [9, 10, 11]. These studies do not establish event-level early warning of connected-to-fragmented FANET transitions. They also commonly report randomly mixed samples, which can leak neighbouring states from the same flight into training and testing. The present study instead groups complete runs, freezes thresholds on validation runs, and distinguishes sample classification from transition detection.

The accepted predictive-digital-twin literature demonstrates a higher evidence bar than a stand-alone dashboard: a physical asset, measured sensors, a model, and a validated information flow must be connected [12]. This study does not have a bidirectional hardware connection and therefore calls its visual interface an *offline predictive-twin replay*, not a live digital twin. The terminology prevents a software visualization from being mistaken for deployment evidence.

The research question is: *Can present motion and multi-scale component structure improve short-horizon fragmentation event forecasting under run-grouped evaluation, and what operational claims remain valid when packet, latency, radio-correlation, and real-data checks are added?*

The contributions are:

1. A future-horizon formulation that predicts $\beta_0(t+h)$ and separately scores connected-to-fragmented events with one-to-one alert matching.
2. Kinetic-TopoGuard, combining current state, projected link-margin features, H_0 persistence images, graph summaries, residual count prediction, and calibrated risk scoring.

3. A leakage-controlled evidence suite containing 20 simulation seeds, five-seed neural comparisons, six horizons, a full three-source factorial, temporally correlated radio, physically bounded relay actions, packet-level simulation, and full-path timing.
4. External checks that keep their meanings separate: field-motion sensitivity, AERPAW UAV-to-infrastructure measurements, held-out 60 GHz peer links, and direct three-UAV UWB topology from MILUV [13].
5. A reproducible package whose audit rejects stale caches, missing event metrics, unsupported calibration language, incomplete seed grids, and stale PDFs or archives.

2. Related work

GCN, GAT, and GraphSAGE learn local representations by aggregating neighbouring node features [14, 15, 16]. FANET and non-terrestrial-network research has used such encoders for route selection and reinforcement learning [17, 18]. Temporal graph models add recurrent or continuous-time state, but their performance depends on sequence construction and run-level separation. None of these architectures makes a future component transition its primary event target.

Persistent-homology layers expose global structure that shallow message passing can miss. PH-GNN, PersLay, and related models learn from persistence diagrams or vectorized topological summaries [9, 10, 11]. For H_0 , the finite merge times are exactly represented by minimum-spanning-tree edge lengths. This makes the descriptor computationally attractive for small and medium UAV swarms. The open issue is not whether persistent homology can be computed, but whether it adds predictive information after current connectivity and motion are controlled.

External FANET evidence is fragmented across datasets. Public forestry flights provide real trajectories but no synchronized peer-link labels [19]. AERPAW Datasets 22 and 23 provide aerial cellular KPI and throughput measurements [20, 21]. The WiNES 60 GHz dataset provides measured UAV-to-UAV channel observations [22]. MILUV supplies synchronized motion capture and UWB measurements for three quadcopters [13]. These sources answer different questions and cannot be merged into a single claim of field-validated FANET packet delivery.

3. Method

3.1. Physical graph and forecast target

At time t , UAV positions and velocities are $P_t = \{p_i(t)\}_{i=1}^N$ and $V_t = \{v_i(t)\}_{i=1}^N$. A geometric policy first supplies fixed and adaptive candidate radii. Candidate links are then filtered by a bidirectional received-power rule:

$$(i, j) \in E_t \iff d_{ij}(t) \leq r_t, \quad P_{i \rightarrow j}(t) \geq \Gamma, \quad P_{j \rightarrow i}(t) \geq \Gamma. \quad (1)$$

The link budget uses log-distance path loss, log-normal shadowing, Rayleigh or Nakagami fading, and transmitter/receiver hardware offsets. Hardware offsets are fixed within a run. Shadowing and fading evolve as Gauss–Markov processes, so consecutive radio snapshots are temporally correlated rather than independent draws. The same realization is used for paired fixed/adaptive-policy comparisons.

For a horizon of h ticks, the count target and sample-level risk label are

$$y_t = \beta_0(G_{t+h}), \quad z_t = \mathbb{I}[\beta_0(G_{t+h}) > 1]. \quad (2)$$

The event target is stricter. An event exists only when

$$\beta_0(G_{t-1}) = 1 \quad \wedge \quad \beta_0(G_t) > 1. \quad (3)$$

Thus repeated fragmented samples do not create repeated ground-truth events.

3.2. Motion-conditioned topology features

Constant-velocity projection gives

$$\hat{p}_i(t+h) = p_i(t) + h\Delta t v_i(t). \quad (4)$$

From current and projected positions, the method extracts pair-distance quantiles, minimum margins to the active radius, closing/opening rates, projected component counts, and the number of links expected to cross the radius. Current graph features contain degree, density, clustering, path, and component summaries.

For H_0 persistence, pairwise distances define a Vietoris–Rips filtration. Finite component deaths are the minimum-spanning-tree merge lengths. Gaussian kernels on birth and persistence coordinates form a persistence image π_t . The implementation uses only H_0 ; it does not claim loop or cavity information.

The complete feature vector is

$$x_t = [c_t, g_t, \pi_t, k_t], \quad (5)$$

where c_t is current/lagged component context, g_t is current graph structure, π_t is the persistence image, and k_t contains kinematic projections.

3.3. Residual count and risk models

Kinetic-TopoGuard anchors count prediction at the observed component count:

$$\hat{y}_t = \max\{1, \beta_0(G_t) + \alpha f_\theta(x_t)\}. \quad (6)$$

An ExtraTrees regressor estimates the residual. The scale α is selected on validation runs to minimize MAE and prevents unstable corrections from dominating the physical anchor. A separate balanced ExtraTrees classifier gives $s_t = P(z_t = 1 \mid x_t)$. The alert threshold is selected on validation data only.

For each ground-truth transition, at most one preceding alert in $[t-h, t-1]$ is matched. An alert cannot match multiple events and alerts issued while

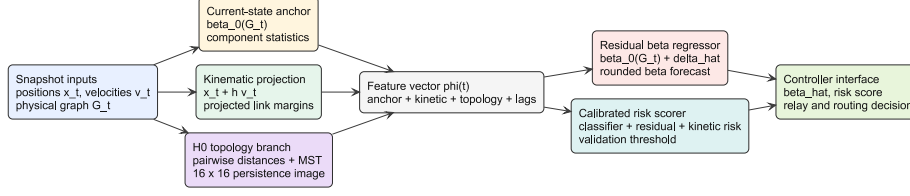


Figure 1: Kinetic-TopoGuard information flow. The controller interface consumes the forecast and risk score, but the reported intervention remains a physically constrained one-step counterfactual rather than a live command path.

Algorithm 1 Kinetic-TopoGuard inference and alerting

Require: P_t, V_t, G_t , horizon h , fitted f_θ, q_ϕ , threshold τ

- 1: Project positions to $\hat{P}_{t+h}$
 - 2: Compute current graph, kinematic, and H_0 persistence features
 - 3: $\hat{y}_t \leftarrow \max(1, \beta_0(G_t) + \alpha f_\theta(x_t))$
 - 4: $s_t \leftarrow q_\phi(x_t)$
 - 5: Emit an alert iff $\beta_0(G_t) = 1$ and $s_t \geq \tau$
 - 6: **return** $\hat{y}_t, s_t$
-

the graph is already fragmented are ignored. The study reports event precision, recall, F_1 , missed events, and unmatched false-alert events per minute. Sample-level F_1 , PR-AUC, ROC-AUC, Brier score, and calibration error remain complementary diagnostics.

3.4. Controller boundary

The forecast can trigger one-step relay steering, route recomputation, or buffering. Relay motion is constrained by current velocity, Δt , maximum speed 30 m s^{-1} , and maximum acceleration 12 m s^{-2} . A relay action receives at most a 3 dB incident-link budget improvement; it does not expand every network edge. The analytical controller output is called a *reachability-delivery proxy*, not PDR. Actual simulated PDR and delay are reported only by the separate SimPy packet experiment. The controller is a counterfactual one-step test and not a flight-control implementation.

4. Experimental protocol

4.1. Simulation and leakage control

Mission trajectories combine scan, transit, and loiter phases. Swarm sizes, mobility families, fixed/adaptive graph policies, and nominal/degraded radio scenarios are crossed within each seed. Complete base runs, identified by mobility, swarm size, and seed, are assigned to train, validation, or test. Policy and radio variants of one trajectory remain in the same split. No test sample selects a learner, residual scale, risk threshold, or alert budget.

The primary confirmation uses 20 independent seeds and compares Kinetic-TopoGuard, validation-selected shallow ML, and current-state persistence. Persistence predicts $\beta_0(t+h) = \beta_0(t)$ and is a deployable no-change baseline. The neural extension uses five seeds for GCN, GAT, GraphSAGE, PI+MLP, two FANET-TopoGNN variants, T-GCN, STGCN, and TGN. Backend records must identify PyTorch; surrogate neural rows are rejected.

MAE and R^2 confidence intervals use clustered bootstrap resampling of complete runs. Risk metrics are computed from held-out samples. Event metrics use the transition matching defined in Section 3. The two operating policies impose validation budgets of two and one false alert events per minute. Feasibility is recorded rather than assumed.

4.2. Stress, ablation, packet, and latency studies

The horizon study reuses each physical trajectory and radio realization and deterministically reindexes labels at 0.2, 0.4, 0.6, 1.0, 1.5, and 2.0 s. Five seeds are used at every horizon. The feature study crosses graph, topology, and kinematic sources in a complete 2^3 factorial. Every row uses the same ExtraTrees learner and current-state context, preventing a stronger classifier from being attributed to one feature family.

The SimPy experiment evaluates 8, 16, 32, and 64 offered packets per 0.1 s tick. Models receive identical source–destination demand and random seeds. It implements finite queues, route availability, CSMA-style random backoff, a single collision domain, and a per-tick deadline. This is a controlled packet abstraction, not ns-3, OMNeT++, or a field MAC trace.

Host-loop timing starts before pairwise-distance construction. It includes fixed/adaptive physical-radio graphs, graph statistics, persistence images, feature assembly, count prediction, and warning score. Five seeds and evenly spread test snapshots are timed. Sensor acquisition, operating-system transport, and autopilot actuation are excluded.

4.3. External evidence layers

Forestry motion. Three synchronized public flight trajectories are sampled at 10 Hz. Because the bags contain no peer RF or packet labels, communication radii are predeclared counterfactual sensitivities. Models and thresholds are frozen from simulation.

AERPAW cellular. LTE connection state is predicted with a chronological final-30% test split and compared with a validation-derived RSRP threshold. iPerf throughput uses both final-30%-by-time and first-half-to-second-half splits. The lag-1 throughput autocorrelation is reported, and interleaved random blocks are prohibited.

60 GHz UAV-to-UAV channel. A measured peer-link model is trained below 33 m and tested on held-out distances at or above 33 m. Link viability uses the same SNR and received-power rule in model testing and downstream sensitivity. Applying the measured distance-success relationship to the forestry

Table 1: Evidence matrix and allowed interpretation.

Source	Directly observed	Evaluation target	Claim boundary
Simulator	Motion and correlated radio are generated	β_0 , transition events	Controlled simulation only
Forestry	Three-UAV field motion	Radius-graph transfer	No measured RF
AERPAW	Cellular KPI and iPerf	Link state, throughput	UAV-to-infrastructure
WiNES 60 GHz	UAV-to-UAV RF measurements	Held-out link viability	Forestry use is transported
MILUV	Three-UAV Vicon and UWB power	UWB topology events	No IP packet delivery

Table 2: Twenty-seed held-out confirmation. Thresholds and operating choices are selected on validation runs; all displayed values are independent-test aggregates.

Model	MAE [95% CI]	R^2 [95% CI]	Sample F_1	Event precision	Event recall	Event F_1	False events/min	PR-AUC
Kinetic-TopoGuard	0.133 [0.122, 0.143]	0.457 [0.376, 0.538]	0.728	0.644	0.390	0.483	3.53	0.838
Shallow ML	0.204 [0.184, 0.224]	0.415 [0.334, 0.497]	0.353	0.380	0.468	0.414	13.24	0.246
Current-state persistence	0.132 [0.122, 0.142]	0.449 [0.369, 0.529]	0.758	0.000	0.000	0.000	0.00	0.695

motion is a *transported RF sensitivity*, not same-site calibration. Primary summaries require all three pair distances to remain inside the measured 6–40 m support; clamped full-trace values are secondary.

MILUV UWB topology. The official Figshare archive for `cir0bstacles_3_random_0` is accessed by HTTP range extraction. Only six Vicon and UWB CSV members are retained, with archive CRC and SHA-256 verification. Motion is interpolated at 10 Hz; reciprocal ranging records are deduplicated; and each robot-pair edge uses the latest causal first-path-power observation no older than 6 s. The longest interval with all three pair states known is evaluated at -90 , -92 , and -95 dBm. No MILUV sample is used for fitting or threshold selection; the three-threshold sensitivity grid is fixed before model evaluation and is not optimized against MILUV outcomes. These labels describe measured UWB ranging quality, not IP PDR.

5. Results and discussion

5.1. Primary confirmation and neural context

Table 2 is generated directly from the 20-seed aggregate. It reports count accuracy together with sample and event measures. The current-state persistence row is ranked as a forecast, not labelled an oracle. Event F_1 is the principal alert measure because it penalizes duplicate and unmatched alerts and exposes missed transitions. Persistence has the lowest MAE (0.132 versus 0.133 for Kinetic-TopoGuard), while Kinetic-TopoGuard has slightly higher R^2 (0.457 versus 0.449). Relative to shallow ML, Kinetic-TopoGuard raises event F_1 from 0.414 to 0.483 and reduces false event alerts from 13.24 to 3.53 per minute. It therefore improves the warning trade-off, not the component-count point estimate.

The five-seed neural extension is retained as model-family context rather than folded into the 20-seed claim. This distinction avoids implying that every expensive neural baseline was run for 20 seeds. Kinetic-TopoGuard obtains MAE 0.120 and event $F_1 = 0.470$. STGCN has the lowest neural MAE (0.218), while TGN has the highest neural event F_1 (0.464) but 6.39 false events/minute

Table 3: Five-seed neural model-family extension.

Model	Seeds	MAE	Sample F_1	Event F_1	Backend
Kinetic-TopoGuard	5	0.120	0.701	0.470	scikit-learn
STGCN (w=5)	5	0.218	0.334	0.404	PyTorch
PI+MLP	5	0.230	0.301	0.448	PyTorch
TGN (w=5)	5	0.234	0.305	0.464	PyTorch
FANET-TopoGNN (concat)	5	0.243	0.318	0.418	PyTorch

Table 4: Five-seed Kinetic-TopoGuard horizon sensitivity.

Horizon (s)	MAE	Sample F_1	Event recall	Event F_1
0.2	0.108	0.740	0.363	0.385
0.4	0.116	0.587	0.445	0.462
0.6	0.120	0.701	0.377	0.470
1.0	0.127	0.596	0.342	0.437
1.5	0.127	0.572	0.269	0.369
2.0	0.132	0.578	0.244	0.355

versus 3.30 for Kinetic-TopoGuard. Backend provenance and per-seed rows are included in the package.

5.2. Horizon and feature-source sensitivity

Table 4 prevents the 600ms result from being interpreted as a universal warning horizon. Longer horizons change both class prevalence and the set of alerts that can match a transition. Event recall and event F_1 are therefore reported alongside MAE and sample F_1 .

The equal-learner factorial in Table 5 is deliberately stricter than a leave-one-feature-out ablation. It shows whether a source helps alone and in interaction with the other sources. The result is not uniformly favourable to topological features: the best source set depends on whether the objective is count MAE, sample risk, or event F_1 . This limits the claim to complementary representation rather than universal dominance.

5.3. Real-data transfer and scope

The forestry transfer tests whether a model trained on simulated mobility remains bounded on measured trajectories. It does not test measured radio. Raw and physically clipped predictions are both archived so that clipping cannot silently create the reported error. The counterfactual graphs contain five, two, and four connected-to-fragmented transitions at the three radii, respectively. All three models obtain zero event recall and event F_1 under their frozen thresholds. Low count MAE and high fragmented-sample F_1 must therefore not be read as successful early warning on this trace.

Table 5: Full 2^3 graph/topology/kinematic factorial; current-state context and learner are fixed in every row.

Feature sources	MAE	Sample F_1	Event F_1
Current state only	0.120	0.460	0.494
Graph	0.121	0.631	0.443
Graph + kinematic	0.120	0.602	0.435
Graph + topology	0.120	0.558	0.436
All sources	0.120	0.575	0.442
Kinematic	0.120	0.646	0.432
Topology	0.120	0.476	0.436
Topology + kinematic	0.120	0.423	0.421

Table 6: Frozen simulation-to-forestry-motion transfer. Link labels are counterfactual radius graphs, not measured radio.

Model	Radius	Frag. rate	MAE	Clip rate	Sample F_1	Event F_1	False events/min
Current-state persistence	q25 (6.3 m)	0.898	0.046 ± 0.000	0.000	0.992 ± 0.000	0.000	0.00
Kinetic-TopoGuard	q25 (6.3 m)	0.898	0.046 ± 0.000	0.000	0.963 ± 0.025	0.000	0.60
Shallow ML	q25 (6.3 m)	0.898	0.882 ± 0.352	0.629	0.636 ± 0.538	0.000	0.75
Current-state persistence	q50 (9.6 m)	0.500	0.021 ± 0.000	0.000	0.994 ± 0.000	0.000	0.00
Kinetic-TopoGuard	q50 (9.6 m)	0.500	0.021 ± 0.000	0.000	0.794 ± 0.175	0.000	0.40
Shallow ML	q50 (9.6 m)	0.500	1.066 ± 0.307	0.462	0.446 ± 0.381	0.000	0.45
Current-state persistence	q75 (13.8 m)	0.292	0.012 ± 0.000	0.000	0.979 ± 0.000	0.000	0.00
Kinetic-TopoGuard	q75 (13.8 m)	0.292	0.012 ± 0.000	0.000	0.738 ± 0.266	0.000	0.55
Shallow ML	q75 (13.8 m)	0.292	1.010 ± 0.560	0.250	0.305 ± 0.254	0.000	0.75

The AERPAW LTE result is strong under chronological holdout: F_1 rises from 0.828 to 0.961 on Dataset 22 and from 0.679 to 0.980 on Dataset 23. Throughput is different. The random forest reduces MAE on the final-30% split but produces a strongly negative R^2 ; on the first-half-to-second-half split it is worse than the training-mean baseline. Together with lag-1 autocorrelation above 0.98, this is negative evidence against the earlier optimistic throughput claim. AERPAW therefore supports cellular link-state transfer, not robust throughput prediction or inter-UAV FANET validation.

The measured 60 GHz logistic model improves held-out long-distance link F_1 from 0.731 to 0.923 and PR-AUC to 0.985. The forestry topology values in Table 8 are reported only as transported sensitivities. Primary rows remain within measured distance support and use the same SNR-plus-received-power definition as the held-out link test.

MILUV closes a different gap by providing simultaneous measured motion and inter-robot UWB topology for three flying vehicles. Table 9 reports frozen-model transfer at three first-path-power thresholds. The small indoor arena and sparse ranging schedule make this a difficult domain shift. Results are interpreted as UWB topology-event evidence, not as outdoor range or packet-routing

Table 7: Chronological AERPAW cellular results.

Evidence	Target/split	Metric	Test n	Baseline	RF/KPI model
AERPAW D22 LTE	Link state/final 30%	F_1	607	0.828	0.961
AERPAW D23 LTE	Link state/final 30%	F_1	690	0.679	0.980
AERPAW D23 iPerf	Final 30%	MAE	362	454.3	397.4
AERPAW D23 iPerf	Final 30%	R^2	362	-16316.103	-13712.894
AERPAW D23 iPerf	First half $\rightarrow$ second half	MAE	603	331.0	458.8
AERPAW D23 iPerf	First half $\rightarrow$ second half	R^2	603	-1.032	-2.966

Table 8: Measured peer-link test and in-support transported forestry sensitivity. For link rows, estimates A/B are baseline/model; for sensitivity rows they are expected β_0 /fragmentation probability.

Evidence	Target	n	Condition	Estimate A	Estimate B
UAV-UAV 60 GHz	Link viability	1756	F_1	0.731	0.923
UAV-UAV 60 GHz	Link viability	1756	PR-AUC	0.576	0.985
Transported RF sensitivity	Expected β_0	1680	SNR 5 dB	1.484	0.402
Transported RF sensitivity	Expected β_0	1680	SNR 7 dB	1.897	0.667
Transported RF sensitivity	Expected β_0	1680	SNR 10 dB	2.325	0.867

performance. At the primary -90 dBm threshold, Kinetic-TopoGuard preserves a low count MAE of 0.045 but reaches only 0.08 event F_1 ; shallow ML reaches 0.40 event F_1 while its count MAE exceeds 1.28. The metric disagreement is evidence against treating either transfer as deployment-ready.

5.4. Operating points, packet behaviour, and latency

Operating points are selected per seed on validation event F_1 subject to one or two false event alerts per minute, then applied once to test runs. If a budget has no feasible threshold, the exported constraint flag records that fact; the analysis does not replace the policy with a test-selected threshold. Both budgets are feasible in all 20 validation seeds. On test runs, the two-event policy attains event $F_1 = 0.421$ at 1.85 false events/minute; the one-event policy trades recall for precision, yielding event $F_1 = 0.351$ at 0.90 false events/minute.

The packet-load sweep in Table 11 separates graph reachability from packet performance. At high offered load, contention-deadline drops dominate and all models degrade. Kinetic-TopoGuard does not consistently outperform the shallow or persistence controller. The correct conclusion is that better warning scores do not automatically translate into better PDR under a constrained MAC and one-step control policy.

Table 12 times the complete host-side topology-to-warning path. Tail latency, not only model inference, is the relevant engineering quantity. The result cannot be extended to onboard execution because sensor transport, scheduling, and autopilot communication are outside the timed boundary. Mean host-loop time rises from 15.81 ms at 10 UAVs to 21.68 ms at 30 UAVs; P95 is 25.68 ms–30.87 ms. The complete path therefore does not satisfy a 20 ms tail-latency claim on this CPU, even though isolated model inference is faster.

Table 9: Frozen transfer to measured MILUV three-UAV UWB topology.

Model	FPP threshold	Events	MAE	Risk PR-AUC	Event F1
Current-state persistence baseline	-95 dBm	1	0.017	0.200	0.000
Kinetic-TopoGuard	-95 dBm	1	0.017	0.018	0.133
Shallow ML	-95 dBm	1	1.266	1.000	0.667
Current-state persistence baseline	-92 dBm	2	0.045	0.034	0.000
Kinetic-TopoGuard	-92 dBm	2	0.045	0.066	0.080
Shallow ML	-92 dBm	2	1.275	0.212	0.400
Current-state persistence baseline	-90 dBm	2	0.045	0.034	0.000
Kinetic-TopoGuard	-90 dBm	2	0.045	0.066	0.080
Shallow ML	-90 dBm	2	1.286	0.212	0.400

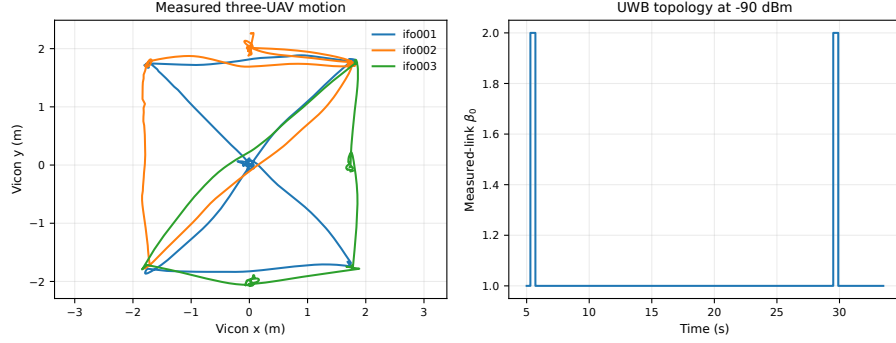


Figure 2: MILUV Vicon trajectories and the causally reconstructed measured-link component trace at the primary first-path-power threshold.

5.5. Comparison with a predictive digital-twin acceptance bar

The bridge-monitoring case study of Jasinski et al. integrates a physical asset, measured sensors, a model-based representation, machine learning, and a validated information flow [12]. The present work now matches several methodological strengths: explicit system architecture, measured-data checks, reproducible model evaluation, and an engineering decision layer. It does not match the physical integration bar. Its replay is generated from completed experiments and sends no command to a UAV. Calling it an offline replay preserves the distinction between predictive analytics and a bidirectional operational twin.

6. Limitations

Four boundaries remain. First, the primary training domain is simulated even though radio errors are temporally correlated and hardware offsets persist within runs. Second, MILUV is indoor UWB ranging quality and contains only three UAVs; it is not outdoor FANET IP traffic. Third, the SimPy model has a single collision domain and simplified routing, queues, and deadlines. Fourth, the relay response is a one-step counterfactual with speed, acceleration, and 3dB link-budget limits, not a closed-loop flight experiment.

Table 10: Validation-selected Kinetic-TopoGuard operating policies on held-out test runs.

Policy	Validation budget	Feasible seeds	τ	Event precision	Event recall	Event F1	False events/min
Deployable	2.0	100%	0.39	0.743	0.303	0.421	1.85
Strict	1.0	100%	0.44	0.816	0.230	0.351	0.90

Table 11: Five-seed SimPy packet results. Delay is computed from delivered-packet samples, not from an average of per-snapshot averages.

Model	Load	PDR	Mean delay	P95 delay	Deadline drop
Current persistence	8	0.969	15.0	28.9	0.000
Kinetic-TopoGuard	8	0.966	14.7	28.0	0.000
Shallow ML	8	0.965	14.7	27.7	0.000
Current persistence	16	0.968	28.3	54.5	0.000
Kinetic-TopoGuard	16	0.965	28.1	53.6	0.000
Shallow ML	16	0.964	27.7	52.7	0.000
Current persistence	32	0.898	50.9	92.8	0.071
Kinetic-TopoGuard	32	0.897	50.8	92.5	0.069
Shallow ML	32	0.908	50.6	91.5	0.056
Current persistence	64	0.406	50.6	95.4	0.562
Kinetic-TopoGuard	64	0.412	50.5	95.4	0.554
Shallow ML	64	0.413	50.7	95.4	0.551

Consequently, no result establishes field PDR, energy use, regulatory feasibility, multi-channel interference, or safe autonomous actuation. The next confirmatory study requires synchronized outdoor peer-to-peer packet traces, onboard timing and power measurement, and a preregistered closed-loop intervention protocol. These are limitations, not claims closed by prose.

7. Reproducibility

The anonymous package contains configurations, source code, per-seed outputs, protocols, derived real-data traces, hashes, generated tables, and figures. The MILUV downloader records the official dataset DOI, archive identifier, archive MD5, member CRC values, and per-file SHA-256 values. A readiness audit checks cache version, seed coverage, PyTorch backend provenance, event columns, validation/test separation, horizon and factorial grids, packet loads, latency stages, support-aware 60 GHz scope, MILUV transfer, PDF page counts, anonymization, ZIP CRC, and package freshness.

The included dashboard is titled “Offline FANET Predictive-Twin Replay.” It binds only to generated CSV/JSON evidence, is interactive for operating-policy comparison, and contains an explicit notice that it is not live, bidirectional, or hardware connected. This interface is a reproducibility aid rather than an independent experiment.

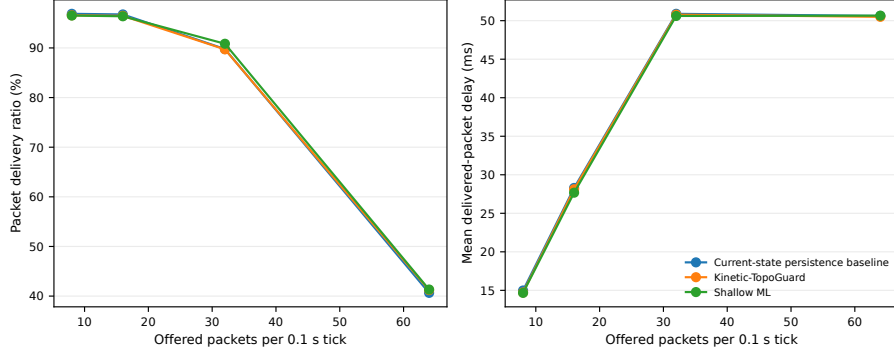


Figure 3: Packet delivery and delivered-packet delay versus offered load.

Table 12: Complete host-loop timing across five seeds.

UAVs	Samples	Mean	P50	P95	P99 (ms)
10	500	15.81	17.57	26.04	35.53
20	500	17.76	19.85	25.68	28.16
30	500	21.68	23.29	30.87	53.42

8. Conclusion

Kinetic-TopoGuard formulates FANET monitoring as future component-count and fragmentation-event forecasting. The revised evaluation replaces ambiguous topology-change alerts with connected-to-fragmented events, uses one-to-one matching, selects thresholds only on validation runs, and adds temporal radio, physical relay limits, factorial, horizon, packet, latency, and real multi-UAV UWB checks. The evidence supports a reproducible short-horizon warning method, but it also shows where claims fail: topological features do not dominate every objective, AERPAW throughput transfer is not robust, and better warning scores do not guarantee higher packet delivery. These negative results define the appropriate boundary for follow-on hardware validation.

Data and code availability

The double-anonymous supplementary package contains the executable code, configurations, derived data, checksums, and result artefacts required for review. Author-identifying archive metadata is withheld from the anonymous manuscript.

CRedit authorship contribution statement

Ercan Erkalkan: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review and editing.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work.

Declaration of generative AI and AI-assisted technologies

Generative AI tools were used for language editing and software-development assistance. The author verified the code, executed experiments, inspected generated artefacts, and retains responsibility for the manuscript.

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